

The toxicity of extracts of plant parts of *Moringa stenopetala* in HEPG2 cells in vitro.

The cytotoxicity of extracts from a widely used species of plant, *Moringa stenopetala*, was assessed in HEPG2 cells, by measuring the leakage of lactate dehydrogenase (LDH) and cell viability. The functional integrity of extract-exposed cells was determined by measuring intracellular levels of ATP and glutathione (GSH). The ethanol extracts of leaves and seeds increased significantly ($p < 0.01$) LDH leakage in a dose- and time-dependent manner. The water extract of leaves and the ethanol extract of the root did not increase LDH leakage. A highly significant ($p < 0.001$) decrease in HEPG2 viability was found after incubating the cells with the highest concentration (500 microg/mL) of the ethanol leaf and seed extracts. At a concentration of 500 microg/mL, the water extract of leaves increased ($p < 0.01$), while the ethanol extract of the same plant part decreased ($p < 0.01$), ATP levels. The root and seed extracts had no significant effect on ATP levels. The ethanol leaf extract decreased GSH levels at a concentration of 500 microg/mL ($p < 0.01$), as did the ethanol extract of the seeds at 250 microg/mL and 500 microg/mL ($p < 0.05$). The water extract of the leaves did not alter GSH or LDH levels or affect cell viability, suggesting that it may be non-toxic, and is consistent with its use as a vegetable. The data obtained from the studies with the ethanol extract of the leaves and seeds from *Moringa stenopetala* show that they contain toxic substances that are extractable with organic solvents or are formed during the process of extraction with these solvents. The significant depletion of ATP and GSH only occurred at concentrations of extract that caused leakage of LDH. Further investigation with this plant in order to identify the constituents extracted and their individual toxic effects both in vivo and in vitro is warranted. This study also illustrates the utility of cell culture for screening plant extracts for potential toxicity. Copyright (c) 2005 John Wiley & Sons, Ltd.